

R2.1 How much? The amount of chemical change

Chemical equations

Learning outcomes

By the end of this section you should be able to deduce chemical equations, including state symbols, when reactants and products are specified.

The chemistry of cooking

A pancake is a flat, circular cake made from flour, eggs and milk and cooked on a griddle or frying pan. You may have tried other similar versions of pancakes from around the world such as dosa, crêpe, blini and okonomiyaki; all of which are the same shape and made with similar ingredients (**Figure 1**).

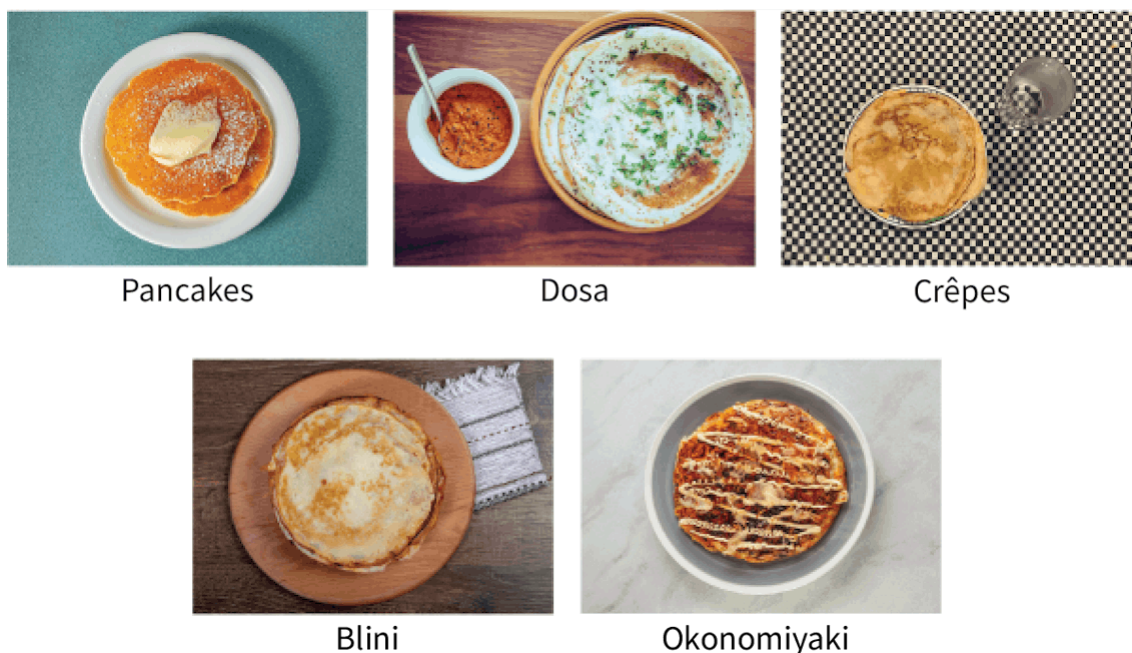


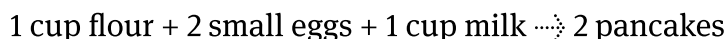
Figure 1. Pancakes, dosa, crêpe, blini and okonomiyaki.

Credits: Chris Schneider / 500px, Getty Images, GCShutter, Getty Images, Andres Jabois,

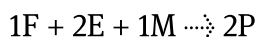
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The ingredients to make two western-style pancakes are one cup of flour, two small eggs and one cup of milk. This list of ingredients could be rewritten in a similar format to how a chemical reaction is represented:



Notice how the quantity of each ingredient is represented by using a number placed before it. For convenience, a chef could also choose to abbreviate the ingredients and use a letter to represent each food:



What are the benefits of choosing to represent a recipe in this way?

Chemical equations

Chemical reactions are written similarly to the pancake recipe. The names of the reactants and products make up the word equation, but the chemical equation is written using the symbols from the periodic table which make up their chemical formula.

Study skills

When writing the symbols of elements from the periodic table be careful to make a clear distinction between the upper-case and lower-case letters. If you write them incorrectly then you will lose the marks in an exam — CO means something quite different to Co.

Making connections

Converting the names of the reactants and products into their chemical formulas is often the part which students struggle with the most. You may find it helpful to recap how to do this before you begin applying the steps above:

- naming ionic compounds (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>)).
- naming molecular compounds (see [subtopic S2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/>)).

- naming organic molecules (see [subtopic S3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)).

Reactants are always written on the left of the arrow and products on the right side. The arrow is important in the equation as it represents a boundary between the reactants and products and also indicates the direction in which the reaction is proceeding. For example, a reaction between potassium metal and water (the reactants) will produce potassium hydroxide and hydrogen gas (the products):



In the equation above you can see that each of the reactants and products are represented using their chemical formulas. Each reactant or product also has a number before it to represent the quantity needed.



In order to make sure that the correct amounts of each reactant and product are present it is important to make sure that the equation is balanced so that the equation observes the law of conservation of mass. You will likely have learned how to balance equations in your previous studies.

Study skills

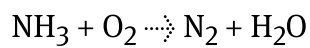
Make sure to always use an arrow when separating the reactants and products (definitely not another symbol such as an equals sign).

You may encounter another variation of this arrow in [section R 2.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/>), known as a 'double arrow' ($\rightleftharpoons$), which is used to indicate an equilibrium reaction.

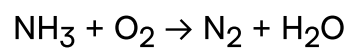
You may also recall other symbols being included above the arrow, such as the words 'pressure' or 'heat'. You may also see the symbol ' ΔH ' (to indicate that energy has been transferred), the chemical formula for a catalyst that has been added or the word 'electricity', to indicate that electrolysis is taking place.

Worked example 1

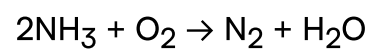
Balance this equation using the simplest whole numbers:



Step 1: Count the number of each element in both sides



	Reactants	
N	1	
H	3	
O	2	

d
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s.S
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p
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F
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n
d
t
h
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l
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w
e
s
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m
o
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m
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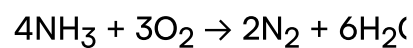
	Reactants	
N	2	
H	3	
O	2	

t,
f
o
r
e
x
a
m
pl
e,
t
h
e
lo
w
e
st
c
o
m
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o
n
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pl
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f
o
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h
i
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S
t
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C
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Remember - do not alter the subscripts.



	Reactants	
N	4	
H	12	

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o
d
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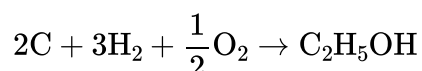
	Reactants	
O	6	

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r
e
a
c
t
a
n
t
s
a
n
d
p
r
o
d
u
c
t
s.

You should have the same number of each element on both sides of the arrow. Remember that you can adjust the coefficients in front of the reactants and products to achieve this, but you cannot change the subscripts. Recall the pancake recipe we previously discussed – when we wanted to specify that the pancake recipe has two eggs in it we wrote it as 2E with the coefficient placed before the eggs. It would be incorrect to write it as E₂ by changing its subscript, as this would indicate that the 2E's are chemically bonded together.

Study skills

As part of subtopic R2.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43479/>) you will have encountered equations which have been balanced using fractions rather than whole numbers. This is particularly important for certain enthalpy changes such as the enthalpy of formation, where the equation must be written to form one mole of a substance. For example, the thermochemical equation used to represent the enthalpy of formation of ethanol is most correctly written as:



It is possible to balance all equations using fractions, but assessment questions will usually have whole numbers so that's what you should practise using.

There are some key features absent from each of the equations above. Consider the pancake recipe we have been discussing.

Option 1: When selecting the ingredients – eggs are in the liquid state when raw and become a solid when cooked. If you have never cooked before, you might be unsure whether this recipe requires raw or cooked eggs.

Option 2: Will the pancakes produced be a solid or a liquid? Providing information about the physical state of the product will make it easier for the chef to understand how long to cook it for and how best to transfer the pancake from the saucepan to its serving plate.

This is also important in chemical reactions. Specifying the physical state of the species involved is relevant as it will determine other features of the reaction, such as how quickly it occurs.

State symbols

Recall your knowledge of the phases of matter from [section S1.1.2a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>). Chemical equations will typically also include the physical state of the reactants and products in equations using state symbols.

For example, the boiling of water could be simply represented by



This is an example using a chemical equation to represent a change in the state of matter, but all other reactions will include them too. The state symbols commonly used are (s) for solid, (l) for liquid, (g) for gas, and (aq) for aqueous, meaning the substance is dissolved in water.

You can see an example of this in **Figure 2**.

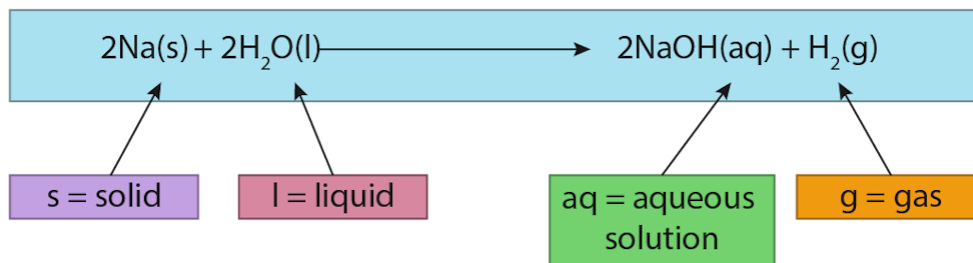


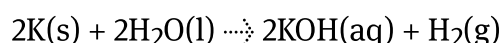
Figure 2. State symbols for a balanced chemical equation.

 More information for figure 2

The image displays a balanced chemical equation: $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$. This equation represents the reaction between sodium and water. Each component of the equation is annotated with state symbols: 's' denotes solid, 'l' depicts liquid, 'aq' stands for aqueous solution, and 'g' indicates gas. These annotations are color-coded and have lines pointing to the respective parts of the chemical equation to clarify the physical state of each reactant and product.

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Try to add state symbols to the potassium and water reaction. You may recall from section S2.1.3b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/>) that ionic compounds are soluble in water.



Why do you think the state symbols are important? This is because they can affect the outcome of a chemical reaction. For example, the rate of a reaction may differ depending on whether the reactants are in the solid, liquid, or gaseous state.

Also, some reactions can only occur in certain states, such as reactions that occur in aqueous solutions. Therefore, including state symbols in a chemical equation helps to provide a more accurate description of the reaction taking place.

Although there can be exceptions to any 'rules' we have to assign state of matter, you will find the **Table 1** useful in helping you determine the state of matter for each component in the equation.

Table 1. The states of matter of some common elements and

Chemical species	Example	Physical state
Metals	Sodium, Na Magnesium, Mg	Solid
Non-metals	Hydrogen, H ₂ Oxygen, O ₂ Chlorine, Cl ₂ Fluorine, F ₂ Noble gases	Gases
Ionic compounds	Sodium chloride, NaCl Calcium carbonate, CaCO ₃	Solids at room temperature Many dissolve in water, forming aqueous solutions
Molecular compounds	Carbon dioxide, CO ₂ Water, H ₂ O Sucrose, C ₁₂ H ₂₂ O ₁₁	May be solid, liquid or gas depending on intermolecular forces which occur between molecules

Writing a chemical equation

A chemical reaction may be described using a short paragraph. The equation is simply the most universal method of representing this same information concisely. There are a number of steps to follow to help you convert the information into a chemical equation but remember that practice is key.

Steps to write a chemical equation:

1. Write down the word equation, placing reactants on the left of the arrow and products on the right of the arrow.
2. Convert the names of each of the reactants and products into their chemical formulas.
3. Balance the equation.
4. Add state symbols.

Making connections

As part of topic R3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>) you will learn how to write half-equations. These show the movement of electrons in oxidation-reduction reactions and look quite different to a regular balanced chemical equation.

When might it be useful to include half-equations?

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate engagement with issues of global significance
- Demonstrate how to initiate and plan a CAS experience

Chemical reactions have led to significant improvements in the quality of life for many people around the world. One such example is the Haber process  (<https://ourworldindata.org/how-many-people-does-synthetic-fertilizer-feed>) that produces fertiliser which is then used to grow crops. However, these improvements have not been evenly distributed throughout the world and have led to inequalities in areas such as food security  (<https://www.who.int/news/item/06-07-2022-un-report--global-hunger-numbers-rose-to-as-many-as-828-million-in-2021>).

What can you do as an IB student to help those less fortunate? Possible CAS experiences could be to organise a food drive  (<https://www.feedingamerica.org/ways-to-give/food-drives>) or to support a charity in your local community.

The next activity will allow you a chance to practise converting different descriptions and videos of reactions into balanced chemical equations.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

Convert these word equations into balanced chemical equations. You should first complete this on your own and then compare your equations to your partner. Some of the equations will include information that is not required. Figuring out what should be included is key!

1. Zinc and lead(II) nitrate react to produce zinc nitrate and lead.
2. Aluminium bromide and chlorine react to produce aluminium chloride and bromine.
3. Sodium phosphate and calcium chloride will react to yield a precipitate of calcium phosphate and aqueous sodium chloride.
4. Aluminium metal reacts with hydrochloric acid to produce hydrogen gas and aluminium chloride.
5. Dilute hydrochloric acid was added to calcium carbonate powder. Bubbles of carbon dioxide, water and calcium chloride solution were formed.
Watch **Video 1** to help you create the balanced chemical equation for this reaction.

Double Displacement Calcium Carbonate and HCl



Video 1. Reaction of calcium carbonate and hydrochloric acid.

6. A reaction between calcium hydroxide and phosphoric acid produces a solution composed of calcium phosphate and water.
7. In a reaction between copper metal and sulfuric acid, a blue solution was formed composed of copper(II) sulfate, water and bubbles of sulfur dioxide.
8. Hydrogen and nitrogen monoxide, two colourless gases, react together and yield bubbles of nitrogen gas and water.
9. A reaction occurs between sodium and hydrochloric acid to produce bubbles of hydrogen and sodium chloride. Watch **Video 2** to help you create the balanced chemical equation for this reaction.

Sodium metal reacting with concentrated hydrochl...**Video 2. Reaction of sodium and hydrochloric acid.**

 More information for video 2

1
00:00:00,534 --> 00:00:02,236
[background noise]
2
00:00:02,302 --> 00:00:05,906
Instructor: Here's the reaction of sodium
with concentrated hydrochloric acid,
3
00:00:06,406 --> 00:00:08,008
12 molar hydrochloric acid.
4
00:00:09,209 --> 00:00:12,045
Sodium metal,
I have it sitting on the counter

5

00:00:13,247 --> 00:00:17,117

and I'm going to add
five milliliters of 12 molar

6

00:00:17,384 --> 00:00:21,455

concentrated hydrochloric
acid to the beaker.

7

00:00:21,989 --> 00:00:25,359

Now I'm going to cut
a fresh piece of sodium.

8

00:00:29,763 --> 00:00:32,199

It's a very soft metal, cuts very easily.

9

00:00:40,674 --> 00:00:44,578

I'm going to add the sodium metal
to the hydrochloric acid.

Reacting ratios

Learning outcomes

By the end of this section you should be able to:

- Calculate reacting masses and/or volumes of reactants and products.
- Calculate the concentrations of reactants and products for reactions occurring in solution.

Recall the recipe for pancakes that we discussed previously where one cup of flour, two small eggs and one cup of milk was mixed and cooked to produce two pancakes. We represented this recipe as the chemical equation as shown in **Figure 1**.

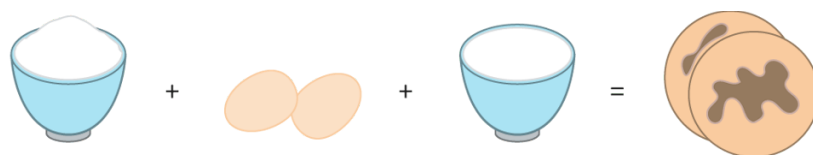
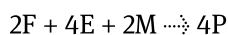
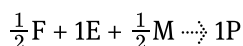


Figure 1. Pancake recipe equation.

What would need to be done to the recipe to make four pancakes? You're right, we could double it. This would mean that you would have two cups of flour, four small eggs and two cups of milk. This could then be represented by:



You can see that the ratio of ingredients has not changed. What if you had lots of flour and milk and one egg in total. How many pancakes could you make?



One pancake! If you tried to use a full cup of flour or milk, then the food produced would certainly not taste like a pancake!

We can figure out how much product can be made by looking at the amount of ingredients we have, or we can figure out the amount of ingredients we will need based on the number of pancakes we want to make (**Figure 2**).

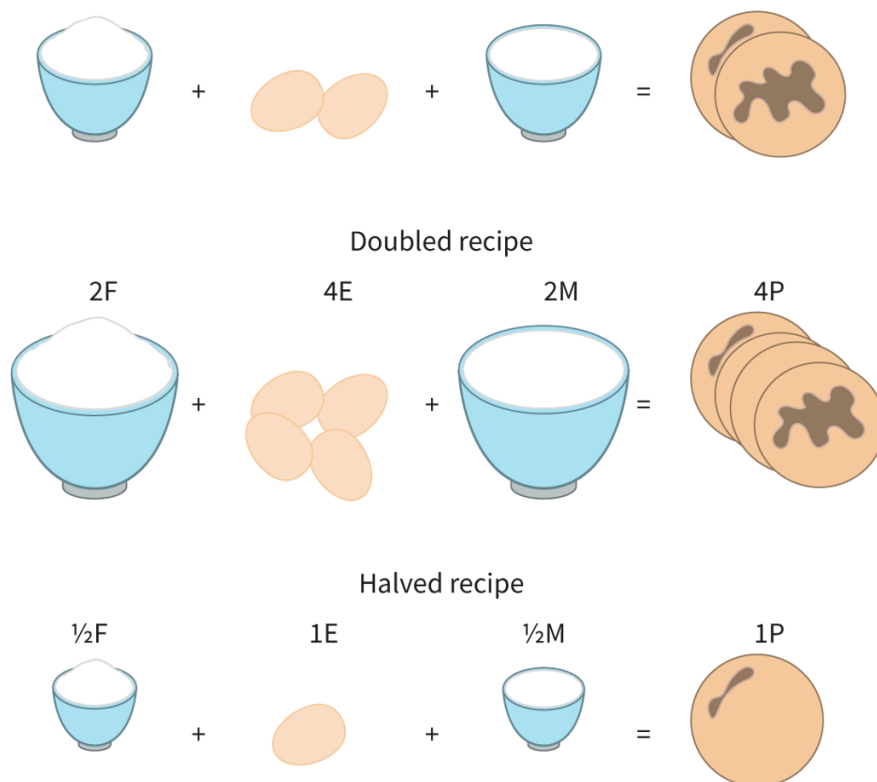


Figure 2. Recipes can be scaled up or down so long as the ratios stay the same.

More information for figure 2

The image consists of three parts that show a basic recipe, a doubled recipe, and a halved recipe. Each section uses illustrated icons of ingredients, which include bowls labeled with amounts of flour (F) and milk (M), and images of eggs (E), leading to a result in pancakes (P).

1. Basic Recipe:

2. Shows a simple combination of a bowl of flour, two eggs, and a bowl of milk resulting in two pancakes.

3. Doubled Recipe:

4. Illustrates twice the amount, with 2F (double the flour), 4E (four eggs), and 2M (double the milk), which produces 4P (four pancakes).

5. Halved Recipe:

6. Displays half the amount required, with $\frac{1}{2}$ F (half the flour), 1E (one egg), and $\frac{1}{2}$ M (half the milk), producing 1P (one pancake).

Each variation is aligned horizontally with simple mathematical symbols to emphasize the scaling, showing that ingredient ratios must remain constant to maintain the recipe's consistency.

[Generated by AI]

This same idea can be applied to chemical equations. You can use the ratios in the reaction to determine how much of each reactant is needed or how much product can be created. Similar to how it is important to make sure that a recipe is correct to begin with, it is important to make sure that chemical equations are always balanced before you begin. If this is something you struggle with, you may want to revise [subtopic S1.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/) before you begin.

Reacting ratios

When we write balanced chemical equations, we place coefficients in front of reactants and products. What do these numbers represent? As long as the correct ratios are maintained, we can actually apply any unit of measurement to them.

Take the reaction between nitrogen and hydrogen gases to produce ammonia molecules for example (**Figure 3**).

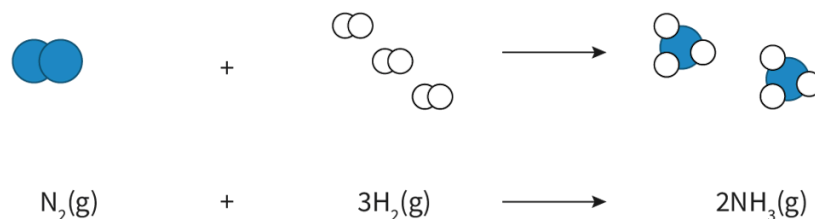


Figure 3. The reaction between nitrogen and hydrogen gases to produce ammonia molecules.

 More information for figure 3

This diagram illustrates the chemical reaction between nitrogen (N_2) and hydrogen (H_2) gases to form ammonia (NH_3). On the left side, there is a depiction of one nitrogen molecule (N_2), represented as two blue circles connected, indicating a diatomic nitrogen gas. Next to it, three hydrogen molecules (H_2) are shown as pairs of smaller white circles. An arrow points from these reactants to the right side, illustrating the formation of ammonia molecules (NH_3). Two ammonia molecules are depicted as structures with a central blue circle (indicating nitrogen) surrounded by three smaller white circles (indicating hydrogen). Below this visual representation, the equation is expressed as: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$.

[Generated by AI]

The ratios could represent the number of molecules (**Figure 4**):

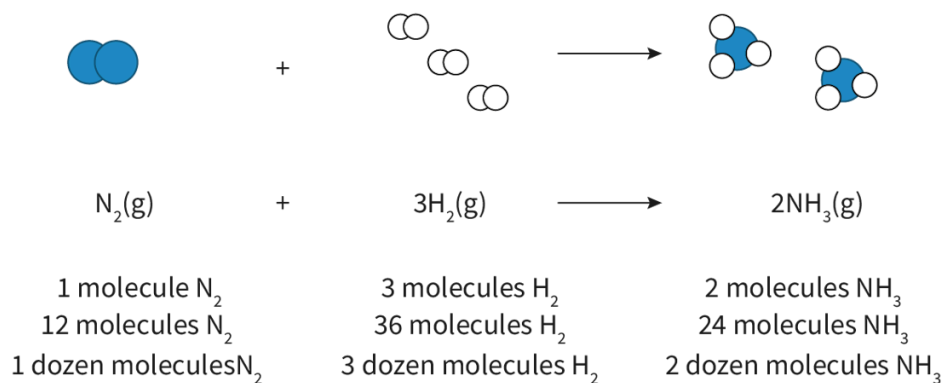


Figure 4. The reaction between nitrogen and hydrogen gases to produce ammonia represented using molecules.

More information for figure 4

The image illustrates the chemical reaction between nitrogen (N_2) and hydrogen (H_2) gases to produce ammonia (NH_3). It depicts molecules of nitrogen and hydrogen combining to form ammonia molecules. On the left, the image shows a pair of blue circles labeled as one molecule of nitrogen (N_2). Adjacent to it are three pairs of white circles representing three molecules of hydrogen (H_2). An arrow points towards the right signifying the reaction, leading to the creation of two ammonia molecules, represented by blue and white linked circles. Below the diagram are proportional representations: 1 molecule N_2 , 3 molecules H_2 , resulting in 2 molecules NH_3 ; 12 molecules N_2 , 36 molecules H_2 , resulting in 24 molecules NH_3 ; 1 dozen molecules N_2 , 3 dozen molecules H_2 , producing 2 dozen molecules NH_3 .

[Generated by AI]

However, trying to measure out the number of molecules would be quite difficult. One hydrogen molecule has a mass of 3.4×10^{-24} g or 0.00000000000000000000000034 g, which is not very easy to measure.

You will recall learning about the mole as a unit of measurement in [subtopic S1.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/). Moles are a convenient way to measure the number of atoms and so the coefficients used to balance chemical equations actually represent moles (**Figure 5**).

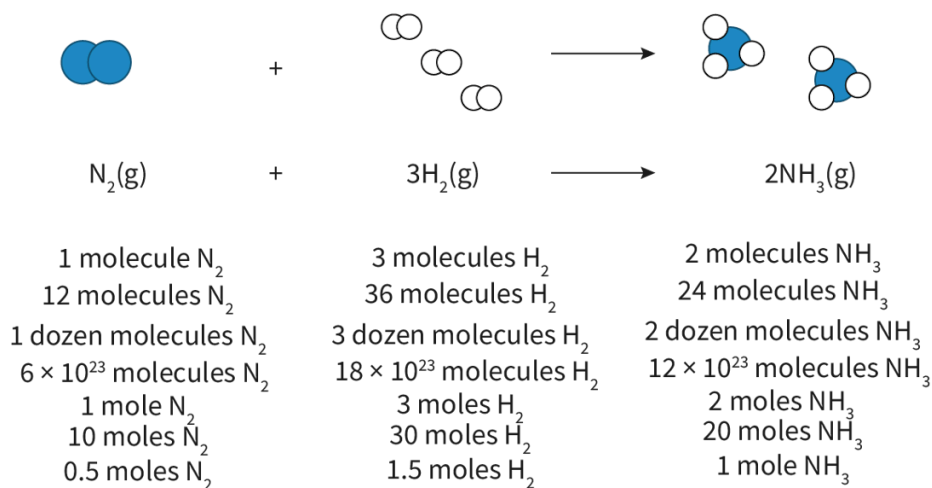


Figure 5. The use of moles to quantify reactants and products.

More information for figure 5

This image is a diagram illustrating a chemical equation involving nitrogen (N_2), hydrogen (H_2), and ammonia (NH_3). It visually represents the reaction of nitrogen gas with hydrogen gas to produce ammonia gas. The left side of the equation shows 1 molecule of N_2 combining with 3 molecules of H_2 . Different representations of these molecules are included: molecules, dozens of molecules, Avogadro's number (6×10^{23} and its multiples), and moles. The right side of the equation displays the products, 2 molecules of NH_3 , along with their equivalent representations including dozens, Avogadro's number, and moles. Arrows indicate the direction of the chemical reaction from reactants to products.

[Generated by AI]

You may find it helpful to practise converting mole ratios in a chemical equation before you continue, and don't forget what you learned about Avogadro's law of combining volumes in [section S1.4.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/).

Worked example 1 shows this in practice.

Worked example 1

Each reaction has a specific mole ratio.

Deduce the missing of the coefficients when the amount of reactant changes.

Question 1

Chemical Equation	4Na	+	O ₂	→	2Na
Mole Ratio	24.0 mol				

(mol)	0.100 mol				
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Question 2

Chemical Equation	4NH ₃	+	3O ₂	→	2N ₂	+	6H ₂ O
Mole Ratio (mol)	0.500 mol						
	10.0 mol						

Question 1

S o l u t i o n s t e p s	Calculations											
S t e p 1: I d e n t i f y t h e s i m p l e s t w h o l	<table><tr><td>Chemical Equation</td><td>4Na</td><td>+</td><td>O₂</td></tr><tr><td>Mole Ratio (mol)</td><td>4</td><td>:</td><td>1</td></tr></table>				Chemical Equation	4Na	+	O ₂	Mole Ratio (mol)	4	:	1
Chemical Equation	4Na	+	O ₂									
Mole Ratio (mol)	4	:	1									

S o l u t i o n s t e p s	Calculations			
e n u m b e r r a t i o .				
S t e p 2 : A p p l y t h e r a t i o w i t h n e w c o e f f				
	Chemical Equation	4Na	+	O ₂
	Mole Ratio (mol)	4	:	1
		24.0 mol	:	6.00 mol
		0.100 mol	:	0.025 mol

S o l u t i o n s t e p s	Calculations
ci e n t.	

Question 2

S o l u t i o n s t e p s	Calculations														
S t e p 1: Id e n t i f y t h e s i m p l e s t w h	<table><tr><td>Chemical Equation</td><td>4NH₃</td><td>+</td><td>3O₂</td><td>→</td></tr><tr><td>Mole Ratio (mol)</td><td>4</td><td></td><td>3</td><td></td></tr></table>					Chemical Equation	4NH ₃	+	3O ₂	→	Mole Ratio (mol)	4		3	
Chemical Equation	4NH ₃	+	3O ₂	→											
Mole Ratio (mol)	4		3												

S o l u t i o n s t e p s	Calculations																						
o l e n u m b e r r a t i o .																							
S t e p 2 : A p p l y t h e r a t i o w i t h n e w c o e f	<table><tr><td>Chemical Equation</td><td>4NH₃</td><td>+</td><td>3O₂</td><td>→</td></tr><tr><td rowspan="3">Mole Ratio (mol)</td><td>4</td><td></td><td>3</td><td></td></tr><tr><td>0.500 mol NH₃</td><td>:</td><td>0.375 mol</td><td>:</td></tr><tr><td>10.0 mol NH₃</td><td>:</td><td>7.50 mol</td><td>:</td></tr></table>					Chemical Equation	4NH ₃	+	3O ₂	→	Mole Ratio (mol)	4		3		0.500 mol NH ₃	:	0.375 mol	:	10.0 mol NH ₃	:	7.50 mol	:
Chemical Equation	4NH ₃	+	3O ₂	→																			
Mole Ratio (mol)	4		3																				
	0.500 mol NH ₃	:	0.375 mol	:																			
	10.0 mol NH ₃	:	7.50 mol	:																			

S o l u t i o n s t e p s	Calculations
fi ci e n t.	

Now try to apply this concept in this next question.



Exercise 1

Click a question to answer



1. Fill in the blanks with an appropriate value to reflect the simplest whole number ratio.

Chemical Equation	2Al	+	3Cl ₂	→	2AlCl ₃
Mole Ratio (mol)	2	:	_____	:	2
	5.0 mol	:	_____	:	5.0 mol
	0.10 mol	:	_____	:	0.10 mol

✓ Check answers

Moles and determining mass, concentration or volume

Mole ratios are especially useful as they allow you to easily convert the mole values to a mass, volume or concentration of a particular chemical species, depending on whether it is a solid, liquid, gas or an aqueous solution.

You may recall from [subtopic S1.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/) that the relationship between moles, mass, concentration and volume are represented by formulae can be found in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) of the DP chemistry data booklet. The molar volume of an ideal gas, $22.7 \text{ dm}^3 \text{ mol}^{-1}$, can be found in section 2 (**Figure 6**).

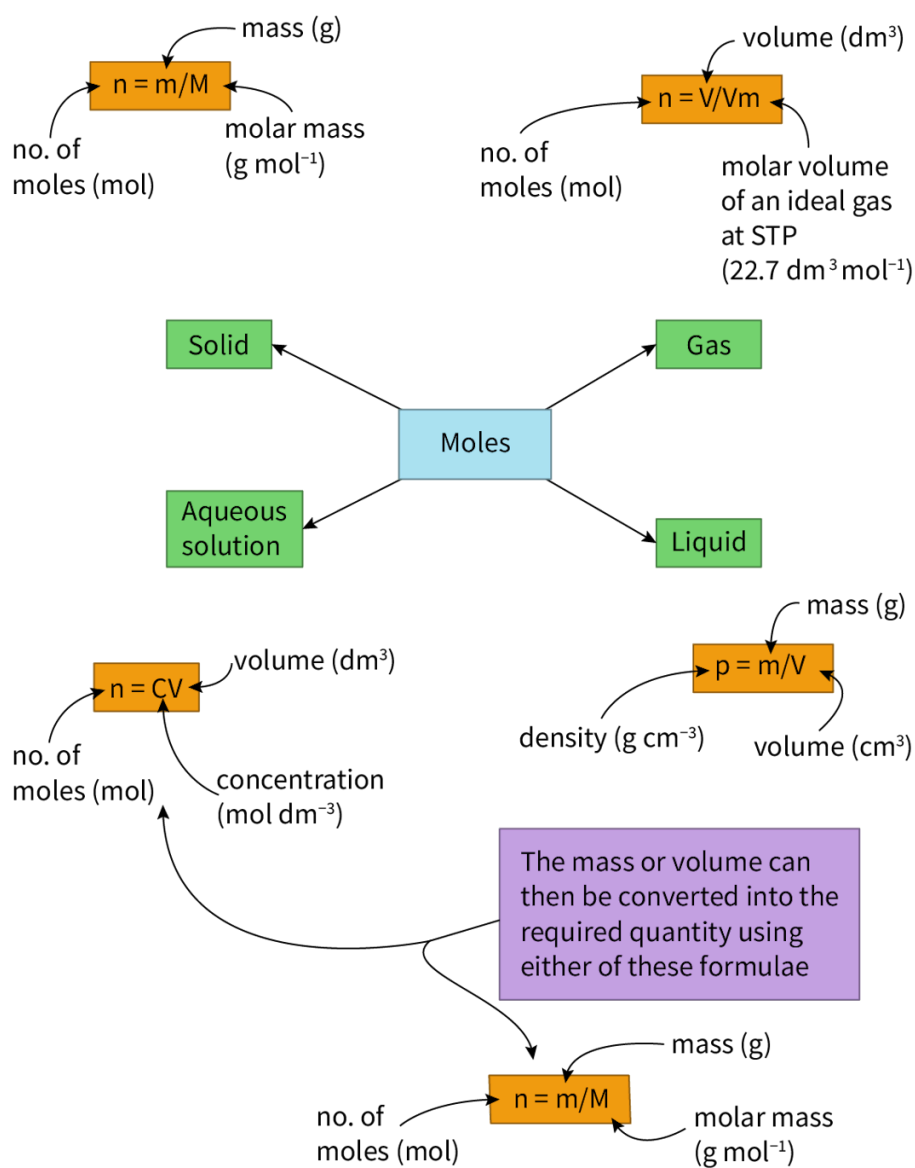


Figure 6. The relationship between moles, mass, concentration and volume.

More information for figure 6

The diagram is a flowchart illustrating the relationships between moles, mass, concentration, and volume. At the center of the diagram, there is a blue box labeled 'Moles.' From this central box, arrows point to and from various boxes:

- To the top-left, a green box labeled 'Solid' is connected to the 'Moles' box.
- Below 'Solid,' an orange box states the formula ' $n = m/M$,' where ' n ' represents the number of moles in moles (mol), ' m ' the mass in grams (g), and ' M ' the molar mass in grams per mole (g mol^{-1}).
- To the top-right, an orange box shows ' $n = V/V_m$,' where ' V ' is volume in cubic decimeters (dm^3), and ' V_m ' is the molar volume of an ideal gas at STP ($22.7 \text{ dm}^3 \text{ mol}^{-1}$).
- Next to it, a green box labeled 'Gas' is also connected to the 'Moles' box.
- A green box labeled 'Liquid' is directly connected 'Moles.'
- Another orange box, ' $\rho = m/V$,' involves ' ρ ' as density in grams per cubic centimeter (g cm^{-3}) coupled with ' m ' mass and ' V ' volume in cubic centimeters (cm^3).
- Finally, an orange box illustrates ' $n = CV$,' leveraging ' C ' concentration in moles per cubic decimeter (mol dm^{-3}) and ' V ' for volume.
- At the bottom right, a purple box notes, 'The mass or volume can then be converted into the required quantity using either of these formulae.'

Overall, the diagram visually explains how different states of matter (solid, liquid, gas) and various formulas connect to the concept of moles.

[Generated by AI]

Practical skills

Tool 1: Experimental techniques — Measuring variables

Inquiry 2: Collecting and processing data — Collecting data

Although they are typically measured by volume, liquids can also be measured using their mass. The moles of liquid are then calculated by using the density of that liquid, and whatever other information is available.

For example, water has a density of 1.000 g cm^{-3} . If a student measured 18.00 g of water then they can identify that it would have a volume of 18.00 cm^3 .

However, not every liquid has a density of 1.000 g cm^{-3} , so volume is not equal to mass for all liquids.

For example, the reaction for sodium and oxygen is given in the worked example above. It is possible to determine the mass of sodium oxide produced from 24.0 moles of sodium.

24.0 moles of sodium react with 6.00 moles of oxygen to produce 12.0 moles of sodium oxide. What is the mass of 12.0 moles of sodium oxide, Na_2O ?

Chemical Equation	4Na	+	O_2	$\rightarrow$	$2\text{Na}_2\text{O}$
Mole Ratio (mol)	24.0 mol	:	6.00 mol	:	12.0 mol

Determine the M of Na_2O using the periodic table and input the values into the formula (**Figure 7**).

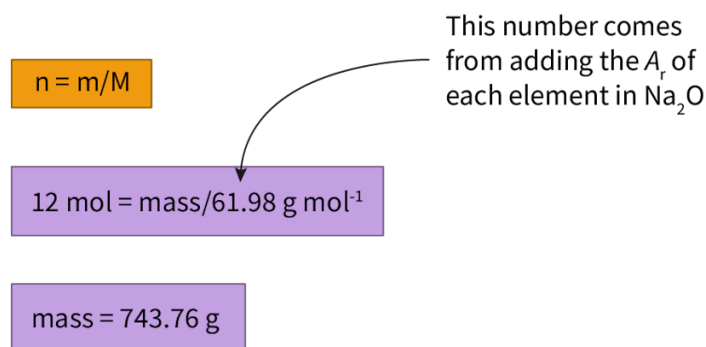


Figure 7. Finding the mass of sodium oxide.

 More information for figure 7

The diagram illustrates the calculation of the mass of sodium oxide using a formula. It starts with a formula box labeled ' $n = m/M$ ' in orange, where ' n ' is moles, ' m ' is mass, and ' M ' is molar mass. Below this, a purple box shows the equation ' $12 \text{ mol} = \text{mass} / 61.98 \text{ g mol}^{-1}$ ', where 61.98 g mol^{-1} is the molar mass derived from adding the relative atomic mass (A_r) of each element in Na_2O . The molar mass calculation is explained in a text box pointing to the equation, stating, 'This number comes from adding the A_r of each element in Na_2O '. Below this, another purple box shows ' $\text{mass} = 743.76 \text{ g}$ ', indicating that the mass is 743.76 grams after multiplying the molar mass by the moles.

[Generated by AI]

What would happen if you were not given the moles of sodium above but were instead given a mass of sodium?

Of course, you could just convert the mass into moles using the same formula.

Study skills

The values for A_r given in [section 7 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) of the DP chemistry data booklet are to two decimal places and so these should be used in calculations.

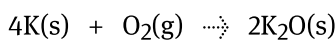
In the lab we always measure in mass, or volume and concentration so you can expect to have to first convert this into moles before you can apply the mole ratio. The following steps summarise how to determine the mass, volume or concentration of an unknown reactant or product:

1. Balance the equation. Identify the known reactant/product and unknown reactant/product.
2. Convert the 'known' mass, concentration or volume given the question into moles.
3. Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.
4. Convert the moles value into the required quantity (mass, concentration or volume). Remember – if the question asks for 'amount' of something then it is acceptable to leave the answer in moles.

The next worked example will demonstrate how this can be done for a reaction involving solids.

Worked example 2

Calculate the mass of potassium oxide formed when 7.82 g of potassium is burned in oxygen, according to the following chemical equation.

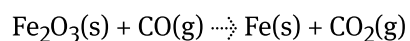


Solution steps	Calculations
Step 1: Identify the 'known' reactant/product and unknown reactant/product.	$4\text{K(s)} + \text{O}_2\text{(g)}$ known
Step 2: Convert the mass, concentration or volume given the question into moles.	For K(s)

Solution steps	Calculations
	Mass, m : 7.82 g $M = 39.10 \text{ g mol}^{-1}$ $n = \frac{m}{M}$ $= \frac{7.82}{39.10}$ $= 0.200 \text{ mol}$
Step 3: Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.	$4\text{K(s)} : 2\text{K}_2\text{CO}_3(\text{s})$ $4 : 2$ 0.200 mol K
Step 4: Convert the moles value into the required quantity (mass, concentration or volume).	For $\text{K}_2\text{O(s)}$ $n = 0.100 \text{ mol}$ $M = 94.20 \text{ g mol}^{-1}$ $m = ?$ $n = \frac{m}{M}$ $0.100 = \frac{m}{94.2}$ $m = 9.42 \text{ g}$

Worked example 3

Calculate the mass (in kg) of iron produced when 32.0 kg of iron(III) oxide is heated with excess CO, according to this unbalanced chemical equation:



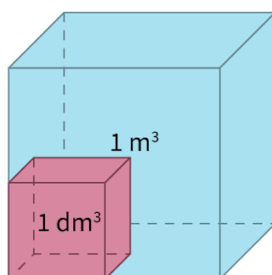
Solution steps	Calculations
Step 1: Balance the equation. Identify the known reactant/product and unknown reactant/product.	$\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g})$ known
Step 2: Convert the mass, concentration or volume given the question into moles. Don't forget to convert units if required, for example, from kg into g.	For $\text{Fe}_2\text{O}_3(\text{s})$ Mass, m : 32.0 kg (note the unit conversion) $M = 159.70 \text{ g mol}^{-1}$ $n = \frac{m}{M}$ $= \frac{32000}{159.70}$ $= 200.4 \text{ mol}$
Step 3: Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.	$\text{Fe}_2\text{O}_3(\text{s}) : 2\text{Fe}(\text{s})$ 1 : 2 200.4 mol Fe_2O_3
Step 4: Convert the moles value into the required quantity (mass, concentration or volume).	For $\text{Fe}(\text{s})$ $n = 400.8 \text{ mol}$ $M = 55.85 \text{ g mol}^{-1}$ $m = ?$ $n = \frac{m}{M}$ $400.8 = \frac{m}{55.85}$ $m = 22\,385 \text{ g of Fe}$ So 22.4 kg of Fe

For a reaction involving aqueous solutions, this process is essentially the same, but now the moles will be converted using $n = CV$ where C is the concentration and V is the volume of the aqueous solution. This is because the quantity of aqueous solution is measured in volume, rather than mass.

Study skills

You learned in [section S1.4.5a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/>) that it is important to be careful with units when dealing with volumes of liquids or gases. You must be able to convert easily between m^3 , dm^3 and cm^3 . This conversion is also given for you in [section 4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/>) of the DP chemistry data booklet.

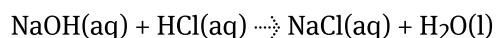
$$\begin{array}{ccccc}
 1 \text{ dm}^3 = 1 \times 10^{-3} \text{ m}^3 & & 1000 \text{ dm}^3 = 1 \text{ m}^3 \\
 1 \text{ cm}^3 = 1 \times 10^{-3} \text{ dm}^3 & & 1000 \text{ cm}^3 = 1 \text{ dm}^3 \\
 \text{Divide by 1000} & & \text{Divide by 1000} \\
 \text{cm}^3 & \xleftrightarrow{\text{Multiply by 1000}} & \text{dm}^3 & \xleftrightarrow{\text{Multiply by 1000}} & \text{m}^3
 \end{array}$$



📖 More information

Worked example 4

A reaction occurs when sodium hydroxide and hydrochloric acid are mixed, as shown in this balanced chemical equation:



23.75 cm^3 of NaOH, with a concentration of $0.01000 \text{ mol dm}^{-3}$, was required to react completely with 25.00 cm^3 of HCl. Determine the concentration of HCl.

Calculations	Calculations
Step 1: Balance the equation. Identify the known reactant/product and unknown reactant/product	$\text{NaOH(aq)} + \text{HCl(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$ known

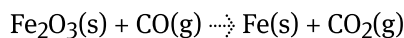
Calculations	Calculations
Step 2: Convert the mass, concentration or volume given the question into moles.	For NaOH(aq) $V = 23.75 \text{ cm}^3$ $= 0.02375 \text{ dm}^3$ (remember to convert) $C = 0.01000 \text{ mol dm}^{-3}$ $n = ?$ $n = CV$ $= 0.01000 \times 0.02375$ $= 0.0002375 \text{ mol}$
Step 3: Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.	NaOH(aq) : HCl(aq) 1 : 1 $0.0002375 : x \text{ mol HCl}$ $x = 0.0002375 \text{ mol}$
Step 4: Convert the moles value into the required quantity (mass, concentration or volume).	For HCl(aq) $n = 0.0002375 \text{ mol}$ $V = 25.00 \text{ cm}^3$ $= 0.025 \text{ dm}^3$ $C = ?$ $n = CV$ $0.0002375 = C \times 0.025$ $C = 0.009500 \text{ mol dm}^{-3}$ $9.500 \times 10^{-3} \text{ mol dm}^{-3}$

The questions above were focused separately on solids and aqueous solutions. However, it is quite normal to have to use a combination of the mole conversion formulas for solids, gases and aqueous solutions to determine the final answer.

Let's revisit the equation for worked example 3 where we calculated the mass of pure iron produced. Carbon dioxide is also produced in that reaction. How could we calculate the volume of carbon dioxide produced? The next worked example explores that, but it would be good practice to try to come up with your own answer before you go through the worked solutions.

Worked example 5

The reaction between 32.00 kg of iron(III) oxide and excess carbon monoxide to produce iron and carbon dioxide was carried out at standard temperature and pressure (STP) with the following unbalanced reaction equation.



Determine the volume of carbon monoxide required for the reaction.

Solution steps	Calculations
Step 1: Balance the equation. Identify the known reactant/product and unknown reactant/product.	$\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightarrow 2\text{Fe}$ unknown
Step 2: Convert the mass, concentration or volume given the question into moles.	For $\text{Fe}_2\text{O}_3(\text{s})$ mass: 32.00 kg = 32 000 (note the unit conversion) $M = 159.70 \text{ g mol}^{-1}$ $n = ?$ $n = \frac{m}{M}$ $= \frac{32000}{159.70}$ $= 200.4 \text{ mol}$
Step 3: Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.	$\text{Fe}_2\text{O}_3(\text{s}) : 3\text{CO}(\text{g})$ 1 : 3 200.4 mol $\text{Fe}_2\text{O}_3 : x \text{ mol}$ $x = 6$
Step 4: Convert the moles value into the required quantity (mass, concentration or volume).	For $\text{CO}(\text{g})$ $n = 601.2 \text{ mol}$ $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ $V = ?$

Solution steps	Calculations
	$n = \frac{V}{V_m}$ $601.2 = \frac{V}{22.7}$ $V = 13\,647 \text{ dm}^3$ $V = 1.365 \times 10^4 \text{ dm}^3 \text{ of C}$

The reaction above required three moles of carbon monoxide which is 68.1 dm^3 of gas. What volume of carbon dioxide would also have been produced from the reaction? Recall what you learned about Avogadro's law in [section S1.4.6](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/>) to help you answer this question.

As both gases have the same coefficient and thus the same number of moles, that means they would have the same volume. Remember that the volume of the gas is independent of what the gas is – it is a molar quantity. However, it is important that the reaction conditions must stay constant as a change in temperature or pressure would impact the volume of the gas.

Worked example 6

What volume of carbon dioxide, CO_2 , in cm^3 , is produced when 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid, HCl , reacts completely with excess calcium carbonate, CaCO_3 ?



Solution steps	Calculations
Step 1: Balance the equation. Identify the known reactant/product and unknown reactant/product.	$\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{a})$ known unknown
Step 2: Convert the mass, concentration or volume given the question into moles.	For $\text{HCl}(\text{aq})$ Volume: $50.0 \text{ cm}^3 = 0.0500 \text{ dm}^3$ $C = 1.00 \text{ mol dm}^{-3}$

Solution steps	Calculations
	$n = ?$ $n = CV$ $n = 1.00 \text{ mol dm}^{-3} \times 0.0500 \text{ dm}^3$
Step 3: Use the mole ratio of a balanced chemical equation to determine the moles of the unknown value.	$\begin{array}{c} \text{HCl(aq)} : \text{CO}_2(\text{g}) \\ 2 : 1 \end{array}$ $\begin{array}{c} 0.0500 \text{ mol HCl} : x \text{ mol CO}_2 \\ x = 0.0250 \text{ mol} \end{array}$
Step 4: Convert the moles value into the required quantity (mass, concentration or volume).	For $\text{CO}_2(\text{g})$ $n = 0.0250 \text{ mol}$ $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ $V = ?$ $n = \frac{V}{V_m}$ $0.0250 = \frac{V}{22.7}$ $V = 0.568 \text{ dm}^3$ $V = 568 \text{ cm}^3 \text{ of CO}_2 \text{ produced}$

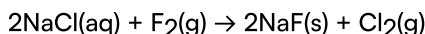
The next activity will allow you an opportunity to practise some of these calculations using experimental data. Remember to consider the physical state of the substances involved before you begin and pay particular attention to units while you complete them.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual activity

Questions

1. Sodium fluoride can be used to treat tooth decay. It is produced in a reaction between fluorine, a pale yellow-green gas, and sodium chloride solution. Chlorine gas is also produced in this reaction, which can be represented by the following balanced equation:



A student wanted to prepare 84.00 g of sodium fluoride according to this reaction.

Calculate the volume of fluorine gas which would be needed to do this.

2. A student wanted to determine the concentration of hydrochloric acid by reacting it with sodium thiosulphate solution. They formulated a balanced chemical equation for the reaction:



16.7 cm³ of a 0.100 mol dm⁻³ sodium thiosulphate solution was required to react with 20.0 cm³ of the hydrochloric acid.

Determine the concentration of the hydrochloric acid solution in mol dm⁻³.

3. Calcium carbonate can be decomposed to produce calcium oxide which is used in the production of glass, cement and agricultural products. The reaction also produces carbon dioxide gas, according to the balanced equation:

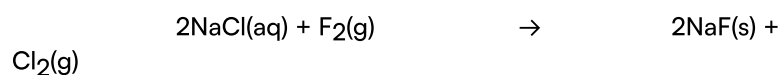


While decomposing calcium carbonate at STP, a student produced 100.00 g of calcium oxide.

Determine the volume of carbon dioxide gas also produced by the reaction in dm³.

Answer 1.

Step 1: Check the equation is balanced

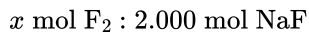
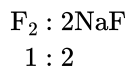


Substance	For F ₂
Property	$V = ?$ $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ $n = ?$

Step 2: Convert the mass into moles of NaF

$$\begin{aligned}
 n &= \frac{m}{M} \\
 &= \frac{84.00}{41.99} \\
 &= 2.000 \text{ mol}
 \end{aligned}$$

Step 3: Use the mole ratio to determine the moles of the unknown value



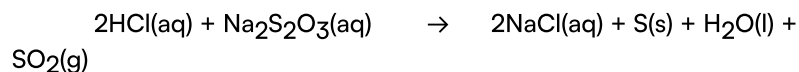
$$x = 1.000 \text{ mol F}_2$$

Step 4: Convert the moles value into volume of F_2

$$\begin{aligned}
 n &= \frac{V}{V_m} \\
 1.000 \text{ mol} &= \frac{V}{22.7} \\
 V &= 22.7 \text{ dm}^3
 \end{aligned}$$

Answer 2.

Step 1: Check the equation is balanced

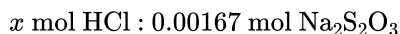
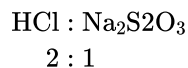


Substance	For HCl	
Property	$n = ?$ $c = ?$ $V = 20.0 \text{ cm}^3$	r $($

Step 2: Convert the concentration and volume into moles of $\text{Na}_2\text{S}_2\text{O}_3$

$$\begin{aligned}
 n &= C \times V \\
 &= 0.100 \times 0.0167 \\
 &= 0.00167 \text{ mol}
 \end{aligned}$$

Step 3: Use the mole ratio to determine the moles of the unknown value



$$x = 0.00334 \text{ mol of HCl}$$

Step 4: Convert the moles value into concentration of HCl

$$\begin{aligned}
 n &= C \times V \\
 3.34 \times 10^{-3} \text{ mol} &= C \times 0.0200 \text{ dm}^3 \\
 C &= 0.167 \text{ mol dm}^{-3}
 \end{aligned}$$

Answer 3.

Step 1: Check the equation is balanced



Substance	For CaO
Property	$n = ?$ $m = 100.00 \text{ g}$ $M = 56.08 \text{ g mol}^{-1}$

Step 2: Convert the mass of CaO into moles.

$$\begin{aligned}
 n &= \frac{m}{M} \\
 &= \frac{100.00}{56.08} \\
 &= 1.783 \text{ mol}
 \end{aligned}$$

Step 3: Use the mole ratio to determine the moles of CO₂ produced

Mole ratio of CaO to CO₂ is 1:1

Amount of CO₂ produced = 1.783 mol

Step 4:

Convert the moles value into volume of CO₂

$$\begin{aligned}
 n &= \frac{V}{V_m} \\
 1.783 \text{ mol} &= \frac{V}{22.7} \\
 V &= 40.5 \text{ dm}^3
 \end{aligned}$$

R2.1 How much? The amount of chemical change

Limiting and excess reactants, theoretical and percentage yield

Learning outcomes

By the end of this section you should be able to:

- (R2.1.3) Identify the limiting and excess reactants from given data.
- (R2.1.3) Distinguish between the theoretical yield and the experimental yield and suggest errors which may cause the experimental yield to be either higher or lower than the theoretical yield.
- (R2.1.4) Solve problems involving reacting quantities, limiting and excess reactants, theoretical, experimental and percentage yields.

Recall the recipe for pancakes we have been discussing and representing as a chemical equation (**Figure 1**).

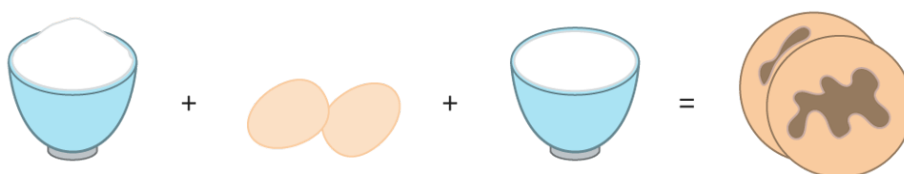
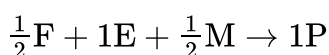


Figure 1. Pancake recipe equation.

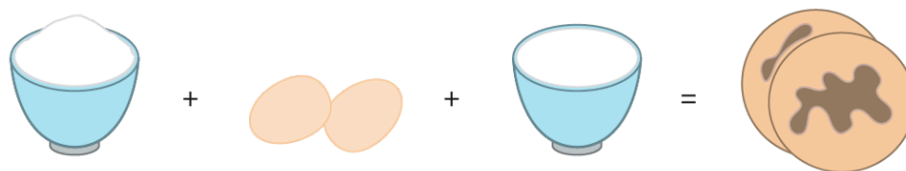
Previously we discussed halving the recipe to make one pancake because there was only one egg:



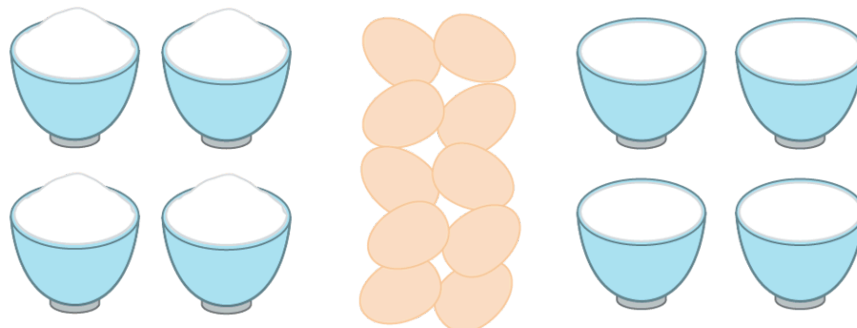
What would happen if there was one egg, but one cup of flour and one cup of milk? Could you make two pancakes?

No, it does not matter if there is excess of other ingredients – the egg will still limit how many pancakes you can make. As the limiting ingredient, the egg decides the number of pancakes you can make, i.e. the yield of pancakes (**Figure 2**).

1 cup of flour + 2 eggs + 1 cup of milk = 2 pancakes



Provided with 4 cups of flour + 10 eggs + 4 cups of milk



We can make 8 pancakes + 2 eggs left over

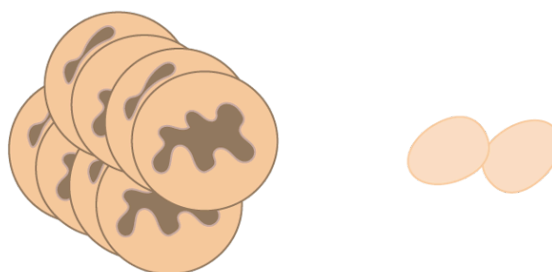


Figure 2. Limiting and excess ingredients.

 More information for figure 2

The image illustrates a recipe scenario to explain the concept of limiting ingredients. At the top, there is a line of text: "1 cup of flour + 2 eggs + 1 cup of milk = 2 pancakes." Below this, three images are arranged horizontally: a bowl of flour, two eggs, and a bowl of milk, with two pancakes at the end, symbolizing the result.

In the middle section, the text reads: "Provided with 4 cups of flour + 10 eggs + 4 cups of milk." This is depicted with four bowls of flour, ten eggs, and four bowls of milk arranged in three rows aligned with their respective text.

At the bottom, the text states: "We can make 8 pancakes + 2 eggs left over." There is an illustration showing eight pancakes stacked together and two leftover eggs, emphasizing that the eggs are the limiting factor in the scenario.

[Generated by AI]

How can we apply this same idea to chemical reactions?

Limiting and excess reactants

Many chemical reactions will also have imperfect amounts of reactants which do not fit the required ratio.

You may have more of one reactant than the other and it simply does not all get used up; it will still remain at the end of the reaction – this reactant is said to be in excess. The limiting reactant is the reactant which is always fully used up. In the example above involving the pancake recipe we identified that the limiting reactant, the egg, was what decided the number of pancakes being produced. This is a key feature of limiting reactants; they always decide the amount of product, known as the yield.

The excess reactant is the one which is remaining after the reaction has taken place. Can you identify the excess reactant in **Figure 3**?

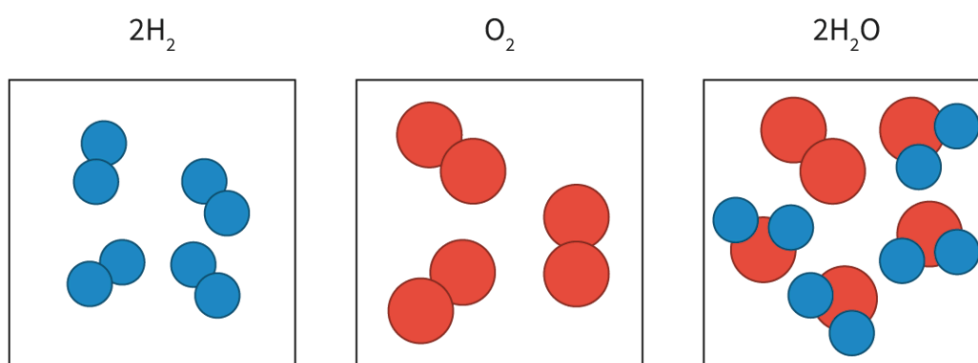


Figure 3. Limiting and excess reactants.

 More information for figure 3

The diagram consists of three sections, each illustrating molecular sets and reactions. The first section displays three pairs of blue circles, each representing molecules of H_2 (hydrogen). The middle section shows two pairs and a single red circle, indicating O_2 (oxygen) molecules. The third section on the right displays molecules forming, showing red and blue circles connected, symbolizing H_2O (water) post-reaction. The image illustrates the transition from reactants to product, with leftover molecules illustrating excess reactant details.

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To determine the limiting reactant, you should:

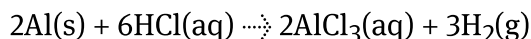
1. Calculate the number of moles of each reactant.
2. Divide the number of moles of each reactant by their coefficient in the reaction. Compare the two numbers – the smaller number is the limiting reactant.
3. Use the original moles of limiting reactant, and the mole ratio to calculate the yield of the product in focus.

The species in the reaction can be present in any phase and you should be prepared to convert solids, liquids and gases into moles using the data you are given.

Worked example 1

10.0 g of aluminium, Al, is added to 50.0 cm³ of a 0.100 mol dm⁻³

solution of hydrochloric acid. Aluminium chloride and hydrogen gas are formed according to this equation:



Determine the limiting reactant.

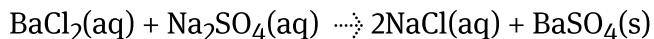
Solution steps	Calculations
Step 1: Determine the limiting reactant by first converting both reactants to moles.	Moles of Al Al mass = 10.0 g Al $M = 26.98 \text{ g mol}^{-1}$ Al moles = $\frac{m}{M}$ = 0.371 mol Moles of HCl HCl volume = 50.0 cm³ = 0.0500 dm³

Solution steps	Calculations
	HCl concentration: 0.100 mol dm⁻³ HCl moles = $C \times V$ = 0.00500 mol
Step 2: Divide the mole values for both by their coefficients.	Mole ratio: 2Al : 6HCl 2 : 6 Moles of Al Al moles: 0.371 mol Coefficient: 2 $\frac{0.371}{2} = 0.186 \text{ mol}$ Moles of HCl HCl moles: 0.00500 mol Coefficient: 6 $\frac{0.005}{6} = 0.000833 \text{ mol}$ The smaller number of moles is this is the limiting reactant.

Once the limiting reactant has been determined it's possible to calculate the yield using the moles and the original mole ratio.

Worked example 2

100.0 cm³ of a 0.5000 mol dm⁻³ solution of BaCl₂ is added to 50.00 cm³ of a 0.1000 mol dm⁻³ solution of Na₂SO₄. A precipitate of BaSO₄ is formed according to this balanced chemical equation:



1. Determine the limiting reactant.
2. Determine the yield of BaSO_4 produced in grams.

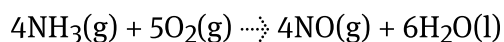
Solution steps	Calculations
Step 1: Determine the limiting reactant by first converting both reactants to moles.	Moles of BaCl_2 $\text{BaCl}_2 \text{ volume} = 100.00 \text{ cm}^3$ $= 0.1000 \text{ dm}^3$ $\text{BaCl}_2 \text{ concentration} = 0.5000$ $\text{BaCl}_2 \text{ moles} = C \times V$ $= 0.05000 \text{ mol}$ Moles of Na_2SO_4 $\text{Na}_2\text{SO}_4 \text{ volume: } 50.00 \text{ cm}^3 =$ $\text{Na}_2\text{SO}_4 \text{ concentration} = 0.10$ $\text{Na}_2\text{SO}_4 \text{ moles: } C \times V = 0.0050$
Step 2: Divide the mole values for both by their coefficients.	Mole ratio $\text{BaCl}_2 : \text{Na}_2\text{SO}_4$ $1 : 1$ Both values are divided by their The smaller number of moles is Na_2SO_4 ($5.000 \times 10^{-3} \text{ mol}$) so
Step 3: Calculate experimental yield using the mole ratio of the limiting reactant and the product in focus.	To calculate theoretical yield of B Mole ratio $\text{Na}_2\text{SO}_4 : \text{BaSO}_4$ $1 : 1$

Solution steps	Calculations
	$5.000 \times 10^{-3} : x$ $x = 5.000 \times 10^{-3} \text{ mol BaSO}_4$ $\text{BaSO}_4 \text{ moles: } 5.000 \times 10^{-3} \text{ mol}$ $\text{BaSO}_4 \text{ } M = 233.40 \text{ g mol}^{-1}$ Mass of BaSO_4 $n = \frac{m}{M}$ $5.000 \times 10^{-3} = \frac{m}{233.40}$ $m = 1.167 \text{ g}$

If you know which reactant is limiting and which one is in excess, then it's also possible to determine how much excess reactant would remain after the reaction.

Worked example 3

A reaction occurs between ammonia and oxygen which forms nitrogen monoxide and water:



100.0 grams of ammonia were reacted with 120.0 grams of oxygen.

(a) Identify which reagent is the limiting reagent.

(b) Calculate the mass of excess reactant which remains when the reaction is completed.

Part (a)

Solution steps	Calculations
Step 1: Determine the limiting reactant by first converting both reactants to moles.	Moles of NH_3 $\text{NH}_3 \text{ mass} = 100.0 \text{ gNH}_3$ $\text{NH}_3 \text{ moles} = \frac{m}{M}$ $= 5.870 \text{ mol}$ Moles of O_2 $\text{O}_2 \text{ mass} = 120.0 \text{ gO}_2$ $\text{O}_2 \text{ moles} = \frac{m}{M}$ $= 3.750 \text{ mol}$
Step 2: Divide the mole values for both by their coefficients.	Mole ratio $4\text{NH}_3 : 5\text{O}_2$ Moles of NH_3 $\text{NH}_3 \text{ moles: } 5.870 \text{ mol}$ Coefficient: 4 $\frac{5.870}{4} = 1.470 \text{ mol}$ Moles of O_2 $\text{O}_2 \text{ moles: } 3.750 \text{ mol}$ Coefficient: 5 $\frac{3.750}{5} = 0.7500 \text{ mol}$ The smaller number of moles is for O_2, so this is the limiting reactant.

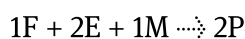
Part (b)

Solution steps	Calculations
Step 1: Calculate the mass of excess reactant which remains after the reaction.	First determine the moles of excess up using mole ratios: Mole ratio $4\text{NH}_3 : 5\text{O}_2$ $x : 3.750$ $x = 3.000 \text{ mol NH}_3 \text{ will be used in the reaction}$ Moles of NH_3 present in 100.00 g of NH_3 $\text{Excess moles of NH}_3 = 5.870 - 3.000$ $= 2.870 \text{ mol}$ So 2.870 mol NH_3 remain after the reaction Mass of excess NH_3 $\text{NH}_3 \text{ moles} = \frac{m}{M}$ $= 2.870 \text{ mol}$ $\text{NH}_3 \text{ } M = 17.04 \text{ g mol}^{-1}$ Hence $\text{NH}_3 \text{ mass} = 48.90 \text{ g}$

Understanding and controlling which reactant is the limiting reactant allows for industry to produce very specific amounts of products. Why do you think the limiting reactant is usually the more expensive one when this is applied in industry?

Theoretical and percentage yield

Consider the original pancake recipe again:



The recipe should yield two pancakes but when pouring the pancake batter, half of it was accidentally spilled. Although there should have been two pancakes yielded in theory, there was actually only one. Half the yield was lost, i.e. 50%.

The theoretical yield that we calculate represents the amount that should be produced if a reaction is fully carried out under ideal conditions. How is theoretical yield calculated?

Based on our understanding that all of the limiting reactant is used up, we can use that to determine what is the maximum product that could be formed from that amount of reactant. The limiting reactant is always what decides the theoretical yield (**Figure 4**).

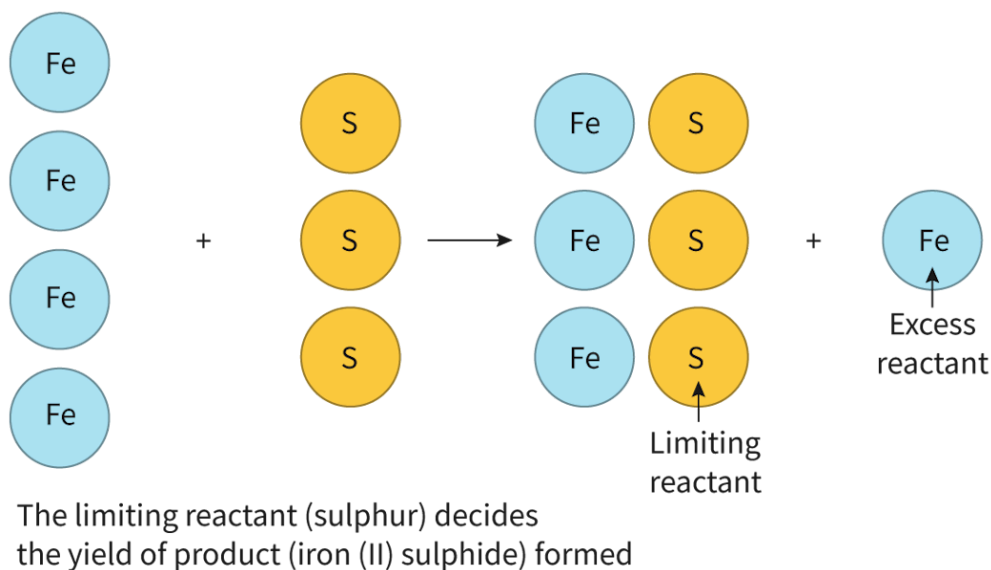


Figure 4. The limiting reactant (sulfur) decides the yield of product (iron(II) sulfide) formed.

 More information for figure 4

The diagram visually represents a chemical reaction between iron (Fe) and sulfur (S) to form iron(II) sulfide. On the left side, there are four blue circles labeled 'Fe' representing iron atoms, and three yellow circles labeled 'S' for sulfur atoms. An arrow points to the reaction products on the right, showing three pairs of Fe and S combined, indicating the formation of iron sulfide. One iron atom is left unreacted, labeled as 'Excess reactant'. A yellow circle marked 'Limiting reactant' near sulfur indicates sulfur is the limiting reactant that decides the yield of the product. This illustrates that the reaction stops when the sulfur is used up.

[Generated by AI]

We can use the limiting reactant to determine the theoretical yield which should be produced but in reality, there are many factors that can cause there to be less product than expected, and result in a lower experimental yield.

Some of these reasons are:

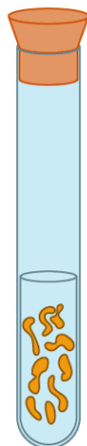
- **Loss of product:** some of the product may evaporate or simply be spilled. It may also be lost when isolating or purifying the product.
- **Incomplete reaction:** there may be impurities present or the reaction may not go to completion due to insufficient time.
- **Alternative reactions:** an unexpected side reaction may occur which results in a different product.
- **One or more of the reactants is impure.**

These things could also occur in the pancake making process. A lower yield obtained could be caused by using flour that was not pure (impurity may undergo a different reaction), not mixing the batter properly (an incomplete reaction), or not scraping clean the batter from the mixing bowl (loss of product).

Making connections

Another reason a reaction may not produce the expected theoretical yield is if the reaction is reversible. If the position of equilibrium lies to the left for that reaction then the reactants will not stay converted into products. You will learn more about this in [subtopic R2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/).

It is also possible for the experimental yield to be greater than the theoretical yield. This can occur when there are impurities present in the product, such as the presence of moisture or a solvent if the drying of a precipitate or solid product is incomplete. It may also occur if the product undergoes subsequent reactions with its surroundings, such as oxidising or decomposing over time (**Figure 5**).



Incomplete drying of a precipitate formed may lead to a higher percentage yield than expected

Figure 5. Incomplete drying can affect the percentage yield.

 More information for figure 5

The image depicts a test tube filled partially with light blue liquid and containing an orange-colored precipitate. The test tube is topped with an orange stopper. Below the test tube, textual information explains that incomplete drying of a precipitate formed may lead to a higher percentage yield than expected. The illustration is intended to demonstrate the concept of incomplete drying affecting chemical yield.

[Generated by AI]

A more convenient way of comparing the theoretical and experimental yield is as a percentage yield. The percentage yield represents the efficiency of the conversion of reactants into products. This is calculated using the formula:

$$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$$

Study skills

The formula for percentage yield is not provided for you in the DP chemistry data booklet so it is important that you can recall it.

Worked example 4

A biochemist was performing a reaction in which she expected to produce 17.89 kg of a product, but actually only produced 11.75 kg of product.

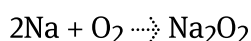
Calculate the percentage yield and suggest reasons why the experimental yield did not achieve 100%.

Solution steps	Calculations
Step 1: Identify theoretical and experimental yield from question.	Theoretical yield = 17.89 kg Experimental yield = 11.75 kg
Step 2: Substitute the values into the equation.	$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$ $= \frac{11.75}{17.89} \times 100\%$ $= 65.68\%$ Possible reasons why experimental yield is less than theoretical yield: • The reaction is reversible so not all reactants are converted to products. • An alternative reaction may have occurred. • Some of the product may have been lost during isolation).

Questions will often require you to first calculate the theoretical yield. You should do this using the value which the limiting reactant would produce.

Worked example 5

Determine the percentage yield for the reaction between 3.74 g of Na and excess O₂ if 5.34 g of Na₂O₂ is recovered:

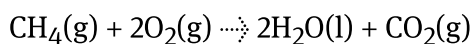


Solution steps	Calculations
Step 1: Determine the limiting reactant and calculate the maximum mass of product which could theoretically be obtained. The question states that oxygen is in excess so sodium must be the limiting reactant.	Moles of Na Na mass = 3.74g Na $M = 22.99 \text{ g mol}^{-1}$ Na moles = $\frac{m}{M}$ = 0.163 mol Mole ratio 2Na : Na₂O₂ 0.163 : $xx = 0.0800 \text{ mol Na}_2\text{O}_2$ Mass of Na₂O₂ Na₂O₂ moles = 0.0815 mol Na₂O₂ $M = 77.98 \text{ g mol}^{-1}$ Mass of Na₂O₂ = $n \times M$ = 6.36 g
Step 2: Identify theoretical and experimental yield.	Theoretical yield = 6.36 g Experimental yield = 5.34 g
Step 3: Substitute the values into the equation.	percentage yield = $\frac{\text{experimental}}{\text{theoretical}}$ = $\frac{5.34}{6.36} \times 100$ = 84.0%

Remember that reactants and products can be solids, liquids, gas or aqueous so be prepared to convert each of them to moles.

Worked example 6

Calculate the percentage yield of water from the combustion of methane if 19.8 g of water was obtained experimentally from a mixture of 20.0 dm³ of methane and 42.0 dm³ of oxygen gas. Assume that the reaction was carried out at STP.



Solution steps	Calculations
Step 1: Determine the limiting reactant and calculate the maximum mass of product which could theoretically be obtained. The question does not specify which reactant is the limiting reactant so that must first be determined.	CH_4 volume = 20.0 dm³ $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ $\text{CH}_4 \text{ moles} = \frac{V}{V_m}$ $= 0.880 \text{ mol}$ O_2 volume = 42.0 dm³ $V_m = 22.7 \text{ dm}^3 \text{ mol}^{-1}$ $\text{O}_2 \text{ moles} = \frac{V}{V_m}$ $= 1.85 \text{ mol}$ Mole ratio is 1 : 2 $\text{CH}_4(\text{g}) : 2\text{O}_2(\text{g})$ If O_2 was the limiting reactant: $x \text{ mol CH}_4 : 1$ $x = 0.930 \text{ mol}$ of CH_4 would be required, but we only have 0.880 mol available. If CH_4 was the limiting reactant: 0.880 mol $\text{CH}_4 : 1.76 \text{ mol O}_2$ $x = 1.76 \text{ mol}$ of O_2 would be required, which is more than the 1.85 mol available. CH_4 must be the limiting reactant and therefore the theoretical yield of the product.

Solution steps	Calculations
Step 2: Identify theoretical and experimental yield.	Experimental yield = 19.8g To calculate theoretical yield of H₂O Mole ratio CH₄ : 2H₂O 1 : 2 0.880 : x $x = 1.76$ mol H₂O moles = 1.76 mol H₂O $M = 18.02$ g mol⁻¹ Mass of H₂O = $n \times M$ = 31.7g Remember that the density of water is 1 g cm⁻³ so the mass and volume for water is quite easy. This (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) booklet.
Step 3: Substitute the values into the equation.	$\text{percentage yield} = \frac{\text{experimental yield}}{\text{theoretical yield}} \times 100\%$ $= \frac{19.8}{31.7} \times 100\%$ $= 62.4\%$

Scientists in industry work to maximise the percentage yield of a reaction to ensure that profits are high and there is very little wastage of raw materials. An example of this is in the pharmaceutical industry where a small difference in percentage purity can translate into millions of dollars.

The next activity will allow you to further explore the concept of limiting and excess reactants and to practise calculating them, the yield, and the percentage yield.

Atom economy

Learning outcomes

By the end of this section you should be able to:

- Calculate the atom economy from the stoichiometry of a reaction using the equation provided for you in the DP chemistry data booklet.
- Discuss the inverse relationship between atom economy and wastage in industrial processes.

Percentage yield is a measure of the amount of product obtained in a reaction compared to what should theoretically be produced according to the stoichiometry of the reaction. A high percentage yield (i.e. close to 100%) indicates that the reaction was very efficient at converting reactants into products while a low percentage yield suggests that some of the starting materials may have been wasted or converted into other unwanted byproducts via an alternative reaction.

Recall the recipe for making pancakes which we have revisited throughout this subtopic. The list of ingredients can be written as a chemical reaction (**Figure 1**).

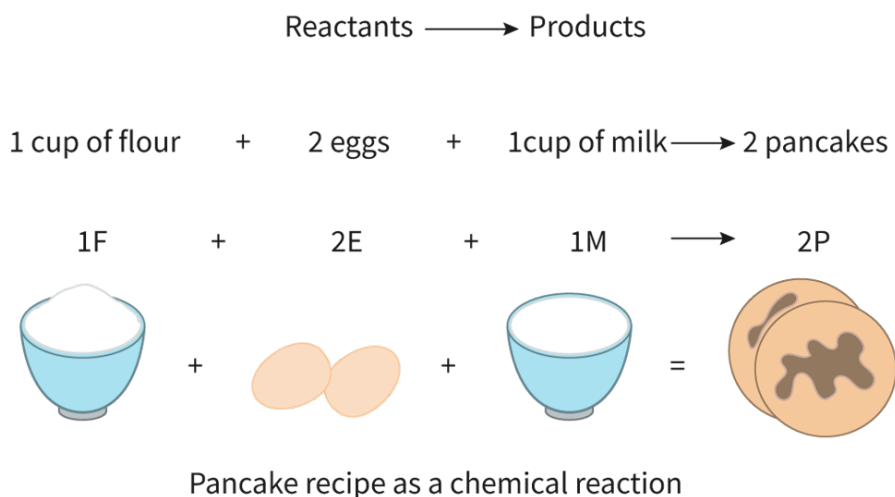


Figure 1. Pancake recipe as a chemical reaction.

 More information for figure 1

This diagram illustrates a pancake recipe represented as a chemical reaction. At the top, there's a label connecting 'Reactants' to 'Products.' Below this, the ingredients are lined up along with their respective chemical symbols. "1 cup of flour" is labeled as "1F,"

"2 eggs" as "2E," and "1 cup of milk" as "1M." These ingredients combine to form the product on the right: "2 pancakes," represented as "2P." Arrows indicate the direction of the reaction from reactants to products. Pictorial representations accompany each ingredient and the pancakes on the far right.

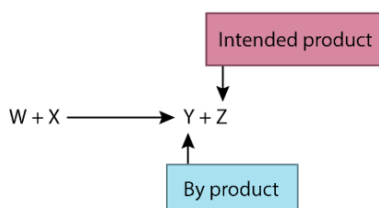
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Chefs have worked hard to ensure that the reactants are pure, are mixed properly, are reacting at the right temperature and no spills occur; as a result they receive a very high percentage yield. But is percentage yield the only metric which matters? If one chef puts eggshells in the compost each time and another chef simply puts them in the trash, should this waste be measured?

Egg shells, empty bags of flour and containers of milk are all disposed of after the reaction. What if there was a way to compost the eggshells or recycle the bags and containers? How could this be quantified?

Evaluating efficiency

Chemical reactions are required to produce substances that do not naturally occur, such as medications, food additives or synthetic polymers. In the synthesis of these types of compounds, the desired product is the focus of the reaction, however there might be other substances formed, known as by-products. For example, a chemist who wants to produce a molecule Z via a reaction between W and X, may also produce molecule Y as a by-product (**Figure 2**).



By-products may also be created in a chemical reaction

Figure 2. By-products may also be created in a chemical reaction.

 More information for figure 2

The image is a flowchart showing the process of a chemical reaction that leads to the formation of an intended product and a by-product. The flowchart consists of two labeled rectangles connected by arrows. The first rectangle at the top is labeled "Intended product," indicating the primary goal of the chemical reaction. Below it, an arrow points downward to the second rectangle labeled "By-product," representing additional substances formed during the reaction. This flowchart visually depicts the concept that chemical reactions can yield both desired and unintended products.

[Generated by AI]

By-products might be useful substances that could be collected and used in another application, or they might serve as waste, causing financial and environmental costs and a challenge for disposal. While percentage yield measures the successful conversion of reactants into the products of the intended reaction, the efficiency of the reaction can also be tracked by another metric: atom economy. This is a measure of the reaction efficiency with respect to the total mass of reactant atoms which end up as desired products, represented as a percentage.

Atom economy is calculated using the total molar masses of reactants and the molar mass of only the desired product:

$$\% \text{ atom economy} = \frac{\text{Molar mass of desired product}}{\text{Molar mass of all reactants}} \times 100$$

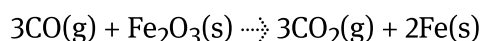
The ideal scenario would be if the atom economy for a reaction was 100%, indicating that all reactants are used to make the desired product with no by-product waste. For two different reactions that make the same desired product using different reactions, the choice where the atom economy is closer to 100% would be the more efficient choice from a green chemistry perspective. The equation is provided for you in [section 1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>) of the DP chemistry data booklet.

To calculate atom economy you should:

1. Calculate the molar mass of the desired product only (including multiplying by the coefficient).
2. Calculate the total molar masses for all the reactants.
3. Substitute values into the equation.

Worked example 1

Iron can be extracted from its ore in the following reaction:



Calculate the atom economy of the reaction.

Solution steps	Calculations
Step 1: Calculate M of the desired product only.	Desired product = iron; $M = 55.85 \text{ g mol}^{-1}$ $\text{Total } M = 2 \times 55.85$ $= 111.70$ (As you are calculating the total M you use the stoichiometric coefficient.)
Step 2: Calculate total M for all the reactants.	$(3 \times M_{\text{C}}) + (3 \times M_{\text{O}}) + (2 \times M_{\text{Fe}})$ $(3 \times 12.01) + (3 \times 16.00) + (2 \times 55.85)$ 243.73
Step 3: Substitute values into the equation.	$\text{Atom economy} = \frac{M \text{ of desired product}}{\text{Total } M \text{ of all reactants}} \times 100\%$ $= \frac{111.70}{243.73} \times 100\%$ $= 45.83\%$

Using the law of conservation of mass we can deduce that no atoms are gained or lost in a chemical reaction. However, some of the atoms may not end up in the desired product, and instead form a by-product. The more by-products you have, the greater the likelihood that you will have a low percentage atom economy.

In the example above, iron is the desired product and carbon dioxide is produced as a by-product, and yet less than one-third of the products are iron according to this metric. You will recall from section R1.3.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fossil-fuels-id-45276/>) that there is a significant relationship between carbon dioxide levels and global warming. This may encourage the industries which are using this reaction to extract iron to explore other reactions which have a greater atom economy and which reduce harmful waste.

Aspect: Global impact of science

The atom economy and the percentage yield both give important information about the 'efficiency' of a chemical process. Think about what else might be needed for a reaction to take place. For example, the energy consumption required in reactions is not measured in either metric; a factor which can have significant global environmental implications. What other metrics do you think might need to be considered?

Atom economy and the amount of waste produced have an inverse relationship. As the waste increases, the percentage atom economy decreases. If a reaction has an atom economy of 31.43%, then 68.57% of the products are waste products.

Practical skills

Tool 1: Experimental techniques — Measuring variables

The ability to identify relationships between variables is an important skill when processing experimental data. Inverse proportionality means that as one variable increases, the other variable decreases, and vice versa. They essentially have an opposite relationship, like when you increase the driving speed, the time it takes to travel that distance decreases.

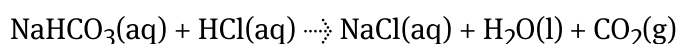
But what if it was possible to find uses for the by-products? For example, the carbon dioxide produced in the above reaction could be sold for use in fire extinguishers or in carbonated beverages. This has both economic and environmental benefits for the scientist carrying out the reaction.

Suggest what the atom economy would be for a reaction which only produces one product?

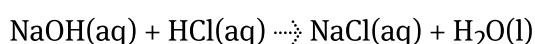
Worked example 2

Sodium chloride, table salt, has many uses in everyday life. There are two reactions which can produce sodium chloride:

Reaction 1



Reaction 2



Use atom economy to determine which reaction is the more efficient reaction.

Reaction 1

Solution steps	Calculations
Step 1: Calculate the M of the desired product only.	Desired product = NaCl; $M = 58.44 \text{ g mol}^{-1}$ $\text{Total } M = 1 \times 58.44 = 58.44$ (Coefficient of NaCl is 1)
Step 2: Calculate total M for all the reactants.	$(1 \times M \text{ Na}) + (1 \times M \text{ H}) + (1 \times M \text{ C}) + (3 \times M \text{ O})$
Step 3: Substitute values into the equation.	$\begin{aligned} \text{Atom economy} &= \frac{M \text{ of desired product}}{\text{Total } M \text{ of all reactants}} \times 100\% \\ &= \frac{58.44}{120.47} \times 100\% \\ &= 48.51\% \end{aligned}$

Reaction 2

Solution steps	Calculations
Step 1: Calculate M of the desired product only.	Desired product = NaCl; $M = 58.44 \text{ g mol}^{-1}$ $\text{Total } M = 1 \times 58.44 = 58.44$ (Coefficient of NaCl is 1)
Step 2: Calculate total M for all the reactants.	All reactants: $(1 \times M \text{ Na}) + (1 \times M \text{ O}) + (1 \times M \text{ H}) + (1 \times M \text{ C})$
Step 3: Substitute values into the equation.	$\begin{aligned} \text{Atom economy} &= \frac{M \text{ of desired product}}{\text{Total } M \text{ of all reactants}} \times 100\% \\ &= \frac{58.44}{76.46} \times 100\% \\ &= 76.43\% \end{aligned}$

Solution steps	Calculations
	The scientists should choose reaction 2 as it is an atom economy of 76.43% compared to rea

A high atom economy indicates that the reaction was efficient at using up all the reactants and producing minimal waste, while a low atom economy suggests that some of the reactants were not used effectively, that unwanted by-products were formed instead or that there was a high amount of waste. This can be helpful in industry when scientists may need to decide how to produce a particular substance.

Theory of Knowledge

The concept of atom economy was first published by Barry Trost in 1991. Industrial chemists are constantly looking at ways to improve the cost and efficiency of making materials at a large scale. While local governments may address regulations for chemical waste, to what extent are chemists responsible for developing new reaction pathways with high atom economies?

Atom economy and green chemistry

Atom economy is one of the key principles of Green Chemistry; a field of science which focuses on creating chemical processes and products that are sustainable and environmentally friendly. It aims to reduce the negative impact of chemistry on our planet and to promote sustainable practices. This often involves reducing the amount of waste products or non-toxic chemicals, simpler reaction pathways and improved efficiency.

Making connections

Recall the impact of a catalyst on the rate of reaction based on your knowledge from subtopic R 2.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43127/>) — catalysts alter the speed of a chemical reaction by providing an alternate pathway for the reaction which has a lower activation energy. How do you think this could be beneficial as one of green chemistry's practices?

You will now have an opportunity to consider the importance of different metrics in measuring reaction efficiency through the format of a debate.

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication skills — Practising active listening skills
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group activity

This activity is based on a debate and so your class will need to be divided into two or more teams. Depending on the size of the class, teams in this debate may be composed of 3-5 students.

Instructions:

1. Choose a team captain. This should be someone who is organised, confident and good at communicating. Their role is to keep the team focused and on track during the debate.
2. Allow 15 minutes research time to prepare the arguments for your team and practice presenting them to your team.
3. The debate should follow the format: Each speaker will have no more than with 90 seconds in total and the team arguing in favour will begin. The opposing team should take responsibility for running the timer for each speaker. Remember to practise good sportsmanship; this is a debate, not a fight!

The motion of the debate is: 'Atom economy is a more important metric than percentage yield'.